F IS LIMIT PRESERVING FOR DIRECTED LIMITS OF NOETHERIAN AFFINE SCHEMES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5635-5640 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 5642-6172, the complete author proof (529 body lines).

Editorial note. Navigation stubs: 26 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 0 of queue_artin.json (pb=22914, stmt 5635-5640, proof 5642-6172).

Editorial note. stacks-artin wave ar01b tail card (2026-10-14): accept per lane-s3/artin_judgments.md row #0; excluded from ar01 as PENDING-CHIEF (raw \cite{ArtinII} at source line 5984); Chief 10:4x ruling -> packaging_substitution path (pointer-class: authorial aside quoting Artin's remark, descent constructed in-span by the four surjectivity sub-blocks); source line 5984 token-replaced to [Artin II], line-count neutral 6590->6590, backup artin-pinned-backup-pre-cite-substitution-20261014.tex, SOURCE.json packaging_substitution+second_pass recorded; zero-cite contract unchanged. Named omissions: a footnote sketch replaces the detailed proof of the etale-local gluing claim, and the functor transformation tilde F -> F is 'constructed in the obvious manner; details omitted' - both construction-bearing steps around them are in-span.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-limit-preserving' also exists in stacks-criteria, stacks-snapshot-20261011 — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00L6: sources/stacks-artin/pinned-original-source.tex, source lines 6172–6172. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BNH: sources/stacks-artin/pinned-original-source.tex, source lines 6172–6172. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:artin.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0GHZ: sources/stacks-artin/pinned-original-source.tex, source lines 6172–6172. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GHE: sources/stacks-artin/pinned-original-source.tex, source lines 5517–5524. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04FH: sources/stacks-artin/pinned-original-source.tex, source lines 5625–5629. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GHB: sources/stacks-artin/pinned-original-source.tex, source lines 5332–5337. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01ZM: sources/stacks-artin/pinned-original-source.tex, source lines 6172–6172. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 081E: sources/stacks-artin/pinned-original-source.tex, source lines 6172–6172. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01Z4: sources/stacks-artin/pinned-original-source.tex, source lines 6172–6172. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01Z0: sources/stacks-artin/pinned-original-source.tex, source lines 6172–6172. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 081F: sources/stacks-artin/pinned-original-source.tex, source lines 6172–6172. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CUF: sources/stacks-artin/pinned-original-source.tex, source lines 6172–6172. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01ZC: sources/stacks-artin/pinned-original-source.tex, source lines 6172–6172. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01YV: sources/stacks-artin/pinned-original-source.tex, source lines 6172–6172. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GHD: sources/stacks-artin/pinned-original-source.tex, source lines 5452–5506. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GC6: sources/stacks-artin/pinned-original-source.tex, source lines 6172–6172. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GI4: sources/stacks-artin/pinned-original-source.tex, source lines 6172–6172. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0ARW: sources/stacks-artin/pinned-original-source.tex, source lines 6172–6172. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0ARB: sources/stacks-artin/pinned-original-source.tex, source lines 6172–6172. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GH8: sources/stacks-artin/pinned-original-source.tex, source lines 5138–5153. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 073K: sources/stacks-artin/pinned-original-source.tex, source lines 6172–6172. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08AS: sources/stacks-artin/pinned-original-source.tex, source lines 6172–6172. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08AZ: sources/stacks-artin/pinned-original-source.tex, source lines 6172–6172. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03XX: sources/stacks-artin/pinned-original-source.tex, source lines 6172–6172. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 049N: sources/stacks-artin/pinned-original-source.tex, source lines 6172–6172. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0A2P: sources/stacks-artin/pinned-original-source.tex, source lines 6172–6172. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Situation 0GH8 the functor F is limit preserving: for any directed limit $V = \lim V_\lambda$ of Noetherian affine schemes over S we have $F(V) = \operatorname{colim} F(V_\lambda)$.*

Proof. This is an absurdly long proof. Much of it consists of standard arguments on limits and étale localization. We urge the reader to skip ahead to the last part of the proof where something interesting happens.

Let $V = \lim_{\lambda \in \Lambda} V_i$ be a directed limit of schemes over S with V and V_λ Noetherian and with affine transition morphisms. See Limits, Section 01YV for material on limits of schemes. We will prove that $\operatorname{colim} F(V_\lambda) \to F(V)$ is bijective.

Proof of injectivity: notation. Let $\lambda \in \Lambda$ and $\xi_{\lambda,1}, \xi_{\lambda,2} \in F(V_\lambda)$ be elements which restrict to the same element of $F(V)$. Write $\xi_{\lambda,1} = (Z_{\lambda,1}, u'_{\lambda,1}, \hat{x}_{\lambda,1})$ and $\xi_{\lambda,2} = (Z_{\lambda,2}, u'_{\lambda,2}, \hat{x}_{\lambda,2})$.

Proof of injectivity: agreement of $Z_{\lambda,i}$. Since $Z_{\lambda,1}$ and $Z_{\lambda,2}$ restrict to the same closed subset of V, we may after increasing i assume $Z_{\lambda,1} = Z_{\lambda,2}$, see Limits, Lemma 0CUF and Topology, Lemma 0A2P. Let us denote the common value $Z_\lambda \subset V_\lambda$, for $\mu \geq \lambda$ denote $Z_\mu \subset V_\mu$ the inverse image in V_μ and and denote Z the inverse image in V. We will use below that $Z = \lim_{\mu \geq \lambda} Z_\mu$ as schemes if we view Z and Z_μ as reduced closed subschemes.

Proof of injectivity: agreement of $u'_{\lambda,i}$. Since U' is locally of finite type over S and since the restrictions of $u'_{\lambda,1}$ and $u'_{\lambda,2}$ to $V \setminus Z$ are the same, we may after increasing λ assume $u'_{\lambda,1} = u'_{\lambda,2}$, see Limits, Proposition 01ZC. Let us denote the common value u'_λ and denote u' the restriction to $V \setminus Z$.

Proof of injectivity: restatement. At this point we have $\xi_{\lambda,1} = (Z_\lambda, u'_\lambda, \hat{x}_{\lambda,1})$ and $\xi_{\lambda,2} = (Z_\lambda, u'_\lambda, \hat{x}_{\lambda,2})$. The main problem we face in this part of the proof is to show that the morphisms $\hat{x}_{\lambda,1}$ and $\hat{x}_{\lambda,2}$ become the same after increasing λ.

Proof of injectivity: agreement of $\hat{x}_{\lambda,i}|_{Z_\lambda}$. Consider the morpisms $\hat{x}_{\lambda,1}|_{Z_\lambda}, \hat{x}_{\lambda,2}|_{Z_\lambda} : Z_\lambda \to W_{red}$. These morphisms restrict to the same morphism $Z \to W_{red}$. Since W_{red} is a scheme locally of finite type over S we see using Limits, Proposition 01ZC that after replacing λ by a bigger index we may assume $\hat{x}_{\lambda,1}|_{Z_\lambda} = \hat{x}_{\lambda,2}|_{Z_\lambda} : Z_\lambda \to W_{red}$.

Proof of injectivity: end. Next, we are going to apply the discussion in Remark 0GHD to V_λ and the two elements $\xi_{\lambda,1}, \xi_{\lambda,2} \in F(V_\lambda)$. This gives us

(1) $e_\lambda : E'_\lambda \to V_\lambda$ separated and locally of finite type,
(2) $e_\lambda^{-1}(V_\lambda \setminus Z_\lambda) \to V_\lambda \setminus Z_\lambda$ is an isomorphism,
(3) a monomorphism $E_{W,\lambda} \to V_{\lambda,/Z_\lambda}$ which is the equalizer of $\hat{x}_{\lambda,1}$ and $\hat{x}_{\lambda,2}$,
(4) a formal modification $E'_{\lambda,/Z_\lambda} \to E_{W,\lambda}$

Assertion (2) holds by assertion (2) in Remark 0GHD and the preparatory work we did above getting $u'_{\lambda,1} = u'_{\lambda,2} = u'_\lambda$. Since $Z_\lambda = (V_{\lambda,/Z_\lambda})_{red}$ factors through $E_{W,\lambda}$ because $\hat{x}_{\lambda,1}|_{Z_\lambda} = \hat{x}_{\lambda,2}|_{Z_\lambda}$ we see from Formal Spaces, Lemma 0GHZ that $E_{W,\lambda} \to V_{\lambda,/Z_\lambda}$ is a closed immersion. Then we see from assertion (4) in Remark 0GHD and Lemma 0GHE applied to the triple E'_λ, $e_\lambda^{-1}(Z_\lambda)$, $E'_{\lambda,/Z_\lambda} \to E_{W,\lambda}$ over V_λ that there exists a closed immersion $E_\lambda \to V_\lambda$ which is a solution for this triple. Next we use assertion (5) in Remark 0GHD which combined with Lemma 0GHB says that E_λ is the "equalizer" of $\xi_{\lambda,1}$ and $\xi_{\lambda,2}$. In particular, we see that $V \to V_\lambda$ factors through E_λ. Then using Limits, Proposition 01ZC once more we find $\mu \geq \lambda$ such that $V_\mu \to V_\lambda$ factors through E_λ and hence the pullbacks of $\xi_{\lambda,1}$ and $\xi_{\lambda,2}$ to V_μ are the same as desired.

Proof of surjectivity: statement. Let $\xi = (Z, u', \hat{x})$ be an element of $F(V)$. We have to find a $\lambda \in \Lambda$ and an element $\xi_\lambda \in F(V_\lambda)$ restricting to ξ.

Proof of surjectivity: the question is étale local. By the unicity proved in the previous part of the proof and by the sheaf property of F in Lemma 04FH, the problem is local on V in the étale topology. More precisely, let $v \in V$. We claim it suffices to find an étale morphism $(\tilde{V}, \tilde{v}) \to (V, v)$ and some λ, some an étale morphism $\tilde{V}_\lambda \to V_\lambda$, and some element $\tilde{\xi}_\lambda \in F(\tilde{V}_\lambda)$ such that $\tilde{V} = \tilde{V}_\lambda \times_{V_\lambda} V$ and $\xi|_U = \tilde{\xi}_\lambda|_U$. We omit a detailed proof of this claim[1].

Proof of surjectivity: rephrasing the problem. Recall that any étale morphism $(\tilde{V}, \tilde{v}) \to (V, v)$ with $\tilde{V}$ affine is the base change of an étale morphism $\tilde{V}_\lambda \to V_\lambda$ with $\tilde{V}_\lambda$ affine for some λ, see for example Topologies, Lemma 049N. Given $\tilde{V}_\lambda$ we have $\tilde{V} = \lim_{\mu \geq \lambda} \tilde{V}_\lambda \times_{V_\lambda} V_\mu$. Hence given $(\tilde{V}, \tilde{v}) \to (V, v)$ étale with $\tilde{V}$ affine, we may replace (V, v) by $(\tilde{V}, \tilde{v})$ and ξ by the restriction of ξ to $\tilde{V}$.

Proof of surjectivity: reduce to base being affine. In particular, suppose $\tilde{S} \subset S$ is an affine open subscheme such that $v \in V$ maps to a point of $\tilde{S}$. Then we may according to the previous paragraph, replace V by $\tilde{V} = \tilde{S} \times_S V$. Of course, if we do this, it suffices to solve the problem for the functor F restricted to the category of locally Noetherian schemes over $\tilde{S}$. This functor is of course the functor associated to the whole situation base changed to $\tilde{S}$. Thus we may and do assume $S = \mathrm{Spec}(R)$ is a Noetherian affine scheme for the rest of the proof.

Proof of surjectivity: easy case. If $v \in V \setminus Z$, then we can take $\tilde{V} = V \setminus Z$. This descends to an open subscheme $\tilde{V}_\lambda \subset V_\lambda$ for some λ by Limits, Lemma 01Z4. Next, after increasing λ we may assume there is a morphism $u'_\lambda : \tilde{V}_\lambda \to U'$ restricting to u'. Taking $\tilde{\xi}_\lambda = (\emptyset, u'_\lambda, \emptyset)$ gives the desired element of $F(\tilde{V}_\lambda)$.

Proof of surjectivity: hard case and reduction to affine W. The most difficult case comes from considering $v \in Z \subset V$. We claim that we can reduce this to the case where W is an affine formal scheme; we urge the reader to skip this argument[2]. Namely, we can choose an étale morphism $\tilde{W} \to W$ where $\tilde{W}$ is an affine formal algebraic space such that the image of v by $\hat{x} : V_{/Z} \to W$ is in the image of $\tilde{W} \to W$ (on reductions). Then the morphisms

$$p : \tilde{W} \times_{W,g} X'_{/T'} \longrightarrow X'_{/T'}$$

and

$$q : \tilde{W} \times_{W,\hat{x}} V_{/Z} \to V_{/Z}$$

are étale morphisms of locally Noetherian formal algebraic spaces. By (an easy case of) Algebraization of Formal Spaces, Theorem 0ARB there exists a morphism $\tilde{X}' \to X'$ of algebraic spaces which is locally of finite type, is an isomorphism over U', and such that $\tilde{X}'_{/T'} \to X'_{/T'}$ is isomorphic to p. By Algebraization of Formal Spaces, Lemma 0GI4 the morphism $\tilde{X}' \to X'$ is étale. Denote $\tilde{T}' \subset |\tilde{X}'|$ the inverse image of T'. Denote $\tilde{U}' \subset \tilde{X}'$ the complementary open subspace. Denote $\tilde{g}' : \tilde{X}'_{/\tilde{T}'} \to \tilde{W}$ the formal modification which is the base change of g by $\tilde{W} \to W$. Then we see that

$$\tilde{X}',\ \tilde{T}',\ \tilde{U}',\ \tilde{W},\ \tilde{g} : \tilde{X}'_{/\tilde{T}'} \to \tilde{W}$$

[1] To prove this one assembles a collection of the morphisms $\tilde{V} \to V$ into a finite étale covering and shows that the corresponding morphisms $\tilde{V}_\lambda \to V_\lambda$ form an étale covering as well (after increasing λ). Next one uses the injectivity to see that the elements $\tilde{\xi}_\lambda$ glue (after increasing λ) and one uses the sheaf property for F to descend these elements to an element of $F(V_\lambda)$.

[2] Artin's approach to the proof of this lemma is to work around this and consequently he can avoid proving the injectivity first. Namely, Artin consistently works with a finite affine étale coverings of all spaces in sight keeping track of the maps between them during the proof. In hindsight that might be preferable to what we do here.

is another example of Situation 0GH8. Denote $\tilde{F}$ the functor constructed from this triple. There is a transformation of functors

$$\tilde{F} \longrightarrow F$$

constructed using the morphisms $\tilde{X}' \to X'$ and $\tilde{W} \to W$ in the obvious manner; details omitted.

Proof of surjectivity: hard case and reduction to affine W, part 2. By the same theorem as used above, there exists a morphism $\tilde{V} \to V$ of algebraic spaces which is locally of finite type, is an isomorphism over $V \setminus Z$ and such that $\tilde{V}_{/Z} \to V_{/Z}$ is isomorphic to q. Denote $\tilde{Z} \subset \tilde{V}$ the inverse image of Z. By Algebraization of Formal Spaces, Lemmas 0GI4 and 0ARW the morphism $\tilde{V} \to V$ is étale and separated. In particular $\tilde{V}$ is a (locally Noetherian) scheme, see for example Morphisms of Spaces, Proposition 03XX. We have the morphism u' which we may view as a morphism

$$\tilde{u}' : \tilde{V} \setminus \tilde{Z} \longrightarrow \tilde{U}'$$

where $\tilde{U}' \subset \tilde{X}'$ is the open mapping isomorphically to U'. We have a morphism

$$\tilde{\hat{x}} : \tilde{V}_{/\tilde{Z}} = \tilde{W} \times_{W,\hat{x}} V_{/Z} \longrightarrow \tilde{W}$$

Namely, here we just use the projection. Thus we have the triple

$$\tilde{\xi} = (\tilde{Z}, \tilde{u}', \tilde{\hat{x}}) \in \tilde{F}(\tilde{V})$$

We omit proving the compatibility condition; hints: if $V' \to V$, $\hat{x}'$, and x' witness the compatibility between u' and $\hat{x}$, then one sets $\tilde{V}' = V' \times_V \tilde{V}$ which comes with morphsms $\tilde{\hat{x}}'$ and $\tilde{x}'$ and show this works. The image of $\tilde{\xi}$ under the transformation $\tilde{F} \to F$ is the restriction of ξ to $\tilde{V}$.

Proof of surjectivity: hard case and reduction to affine W, part 3. By our choice of $\tilde{W} \to W$, there is an affine open $\tilde{V}_{open} \subset \tilde{V}$ (we're running out of notation) whose image in V contains our chosen point $v \in V$. Now by the case studied in the next paragraph and the remarks made earlier, we can descend $\tilde{\xi}|_{\tilde{V}_{open}}$ to some element $\tilde{\xi}_\lambda$ of $\tilde{F}$ over $\tilde{V}_{\lambda,open}$ for some étale morphism $\tilde{V}_{\lambda,open} \to V_\lambda$ whose base change to V is $\tilde{V}_{open}$. Applying the transformation of functors $\tilde{F} \to F$ we obtain the element of $F(\tilde{V}_{\lambda,open})$ we were looking for. This reduces us to the case discussed in the next paragraph.

Proof of surjectivity: the case of an affine W. We have $v \in Z \subset V$ and W is an affine formal algebraic space. Recall that

$$\xi = (Z, u', \hat{x}) \in F(V)$$

We may still replace V by an étale neighbourhood of v. In particular we may and do assume V and V_λ are affine.

Proof of surjectivity: descending Z. We can find a λ and a closed subscheme $Z_\lambda \subset V_\lambda$ such that Z is the base change of Z_λ to V. See Limits, Lemma 01ZM. Warning: we don't know (and in general it won't be true) that Z_λ is a reduced closed subscheme of V_λ. For $\mu \geq \lambda$ denote $Z_\mu \subset V_\mu$ the scheme theoretic inverse image in V_μ. We will use below that $Z = \lim_{\mu \geq \lambda} Z_\mu$ as schemes.

Proof of surjectivity: descending u'. Since U' is locally of finite type over S we may assume after increasing λ that there exists a morphism $u'_\lambda : V_\lambda \setminus Z_\lambda \to U'$ whose restriction to $V \setminus Z$ is u'. See Limits, Proposition 01ZC. For $\mu \geq \lambda$ we will denote u'_μ the restriction of u'_λ to $V_\mu \setminus Z_\mu$.

Proof of surjectivity: descending a witness. Let $V' \to V$, $\hat{x}'$, and x' witness the compatibility between u' and $\hat{x}$. Using the same references as above we may assume (after increasing λ) that there exists a morphism $V'_\lambda \to V_\lambda$ of finite type whose

base change to V is $V' \to V$. After increasing λ we may assume $V'_\lambda \to V_\lambda$ is proper (Limits, Lemma 081F). Next, we may assume $V'_\lambda \to V_\lambda$ is an isomorphism over $V_\lambda \setminus Z_\lambda$ (Limits, Lemma 081E). Next, we may assume there is a morphism $x'_\lambda : V'_\lambda \to X'$ whose restriction to V' is x'. Increasing λ again we may assume x'_λ agrees with u'_λ over $V_\lambda \setminus Z_\lambda$. For $\mu \geq \lambda$ we denote V'_μ and x'_μ the base change of V'_λ and the restriction of x'_λ.

Proof of surjectivity: algebra. Write $W = \mathrm{Spf}(B)$, $V = \mathrm{Spec}(A)$, and for $\mu \geq \lambda$ write $V_\mu = \mathrm{Spec}(A_\mu)$. Denote $I_\mu \subset A_\mu$ and $I \subset A$ the ideals cutting out Z_μ and Z. Then $I_\lambda A_\mu = I_\mu$ and $I_\lambda A = I$. The morphism $\hat{x}$ determines and is determined by a continuous ring map

$$(\hat{x})^\sharp : B \longrightarrow A^\wedge$$

where $A^\wedge$ is the I-adic completion of A. To finish the proof we have to show that this map descends to a map into $A^\wedge_\mu$ for some sufficiently large μ where $A^\wedge_\mu$ is the I_μ-adic completion of A_μ. This is a nontrivial fact; Artin writes in his paper [Artin II]: "Since the data (3.5) involve I-adic completions, which do not commute with direct limits, the verification is somewhat delicate. It is an algebraic analogue of a convergence proof in analysis."

Proof of surjectivity: algebra, more rings. Let us denote

$$C_\mu = \Gamma(V'_\mu, \mathcal{O}) \quad \text{and} \quad C = \Gamma(V', \mathcal{O})$$

Observe that $A \to C$ and $A_\mu \to C_\mu$ are finite ring maps as $V' \to V$ and $V'_\mu \to V_\mu$ are proper morphisms, see Cohomology of Spaces, Lemma 08AS. Since $V = \lim V_\mu$ and $V' = \lim V'_\mu$ we have

$$A = \mathrm{colim}\, A_\mu \quad \text{and} \quad C = \mathrm{colim}\, C_\mu$$

by Limits, Lemma 01Z0[3]. For an element $a \in I$, resp. $a \in I_\mu$ the maps $A_a \to C_a$, resp. $(A_\mu)_a \to (C_\mu)_a$ are isomorphisms by flat base change (Cohomology of Spaces, Lemma 073K). Hence the kernel and cokernel of $A \to C$ is supported on $V(I)$ and similarly for $A_\mu \to C_\mu$. We conclude the kernel and cokernel of $A \to C$ are annihilated by a power of I and the kernel and cokernel of $A_\mu \to C_\mu$ are annihilated by a power of I_μ, see Algebra, Lemma 00L6.

Proof of surjectivity: algebra, more ring maps. Denote $Z_n \subset V$ the nth infinitesimal neighbourhood of Z and denote $Z_{\mu,n} \subset V_\mu$ the nth infinitesimal neighbourhood of Z_μ. By the theorem on formal functions (Cohomology of Spaces, Theorem 08AZ) we have

$$C^\wedge = \lim_n H^0(V' \times_V Z_n, \mathcal{O}) \quad \text{and} \quad C^\wedge_\mu = \lim_n H^0(V'_\mu \times_{V_\mu} Z_{\mu,n}, \mathcal{O})$$

where $C^\wedge$ and $C^\wedge_\mu$ are the completion with respect to I and I_μ. Combining the completion of the morphism $x'_\mu : V'_\mu \to X'$ with the morphism $g : X'_{/T'} \to W$ we obtain

$$g \circ x'_{\mu,/Z_\mu} : V'_{\mu,/Z_\mu} = \mathrm{colim}\, V'_\mu \times_{V_\mu} Z_{\mu,n} \longrightarrow W$$

and hence by the description of the completion $C^\wedge_\mu$ above we obtain a continuous ring homomorphism

$$(g \circ x'_{\mu,/Z_\mu})^\sharp : B \longrightarrow C^\wedge_\mu$$

[3]We don't know that $C_\mu = C_\lambda \otimes_{A_\lambda} A_\mu$ as the various morphisms aren't flat.

The fact that $V' \to V$, $\hat{x}'$, x' witnesses the compatibility between u' and $\hat{x}$ implies the commutativity of the following diagram

$$\begin{array}{ccc} C_\mu^\wedge & \longrightarrow & C^\wedge \\ {\scriptstyle (g \circ x'_{\mu,/Z_\mu})^\sharp}\uparrow & & \uparrow \\ B & \xrightarrow{(\hat{x})^\sharp} & A^\wedge \end{array}$$

Proof of surjectivity: more algebra arguments. Recall that the finite A-modules $\mathrm{Ker}(A \to C)$ and $\mathrm{Coker}(A \to C)$ are annihilated by a power of I and similarly the finite A_μ-modules $\mathrm{Ker}(A_\mu \to C_\mu)$ and $\mathrm{Coker}(A_\mu \to C_\mu)$ are annihilated by a power of I_μ. This implies that these modules are equal to their completions. Since I-adic completion on the category of finite A-modules is exact (see Algebra, Section 0BNH) it follows that we have

$$\mathrm{Coker}(A^\wedge \to C^\wedge) = \mathrm{Coker}(A \to C)$$

and similarly for kernels and for the maps $A_\mu \to C_\mu$. Of course we also have

$$\mathrm{Ker}(A \to C) = \mathrm{colim}\, \mathrm{Ker}(A_\mu \to C_\mu) \quad \text{and} \quad \mathrm{Coker}(A \to C) = \mathrm{colim}\, \mathrm{Coker}(A_\mu \to C_\mu)$$

Recall that $S = \mathrm{Spec}(R)$ is affine. All of the ring maps above are R-algebra homomorphisms as all of the morphisms are morphisms over S. By Algebraization of Formal Spaces, Lemma 0GC6 we see that B is topologically of finite type over R. Say B is topologically generated by $b_1, \ldots, b_n$. Pick some μ (for example λ) and consider the elements

$$\text{images of } (g \circ x'_{\mu,/Z_\mu})^\sharp(b_1), \ldots, (g \circ x'_{\mu,/Z_\mu})^\sharp(b_n) \text{ in } \mathrm{Coker}(A_\mu \to C_\mu)$$

The image of these elements in $\mathrm{Coker}(\alpha)$ are zero by the commutativity of the square above. Since $\mathrm{Coker}(A \to C) = \mathrm{colim}\, \mathrm{Coker}(A_\mu \to C_\mu)$ and these cokernels are equal to their completions we see that after increasing μ we may assume these images are all zero. This means that the continuous homomorphism $(g \circ x'_{\mu,/Z_\mu})^\sharp$ has image contained in $\mathrm{Im}(A_\mu \to C_\mu)$. Choose elements $a_{\mu,j} \in (A_\mu)^\wedge$ mapping to $(g \circ x'_{\mu,/Z_\mu})^\sharp(b_1)$ in $(C_\mu)^\wedge$. Then $a_{\mu,j} \in A_\mu^\wedge$ and $(\hat{x})^\sharp(b_j) \in A^\wedge$ map to the same element of $C^\wedge$ by the commutativity of the square above. Since $\mathrm{Ker}(A \to C) = \mathrm{colim}\, \mathrm{Ker}(A_\mu \to C_\mu)$ and these kernels are equal to their completions, we may after increasing μ adjust our choices of $a_{\mu,j}$ such that the image of $a_{\mu,j}$ in $A^\wedge$ is equal to $(\hat{x})^\sharp(b_j)$.

Proof of surjectivity: final algebra arguments. Let $\mathfrak{b} \subset B$ be the ideal of topologically nilpotent elements. Let $J \subset R[x_1, \ldots, x_n]$ be the ideal consisting of those $h(x_1, \ldots, x_n)$ such that $h(b_1, \ldots, b_n) \in \mathfrak{b}$. Then we get a continuous surjection of topological R-algebras

$$\Phi : R[x_1, \ldots, x_n]^\wedge \longrightarrow B, \quad x_j \longmapsto b_j$$

where the completion on the left hand side is with respect to J. Since $R[x_1, \ldots, x_n]$ is Noetherian we can choose generators $h_1, \ldots, h_m$ for J. By the commutativity of the square above we see that $h_j(a_{\mu,1}, \ldots, a_{\mu,n})$ is an element of $A_\mu^\wedge$ whose image in $A^\wedge$ is contained in $IA^\wedge$. Namely, the ring map $(\hat{x})^\sharp$ is continuous and $IA^\wedge$ is the ideal of topological nilpotent elements of $A^\wedge$ because $A^\wedge/IA^\wedge = A/I$ is reduced. (See Algebra, Section 0BNH for results on completion in Noetherian rings.) Since $A/I = \mathrm{colim}\, A_\mu/I_\mu$ we conclude that after increasing μ we may assume $h_j(a_{\mu,1}, \ldots, a_{\mu,n})$ is in $I_\mu A_\mu^\wedge$. In particular the elements $h_j(a_{\mu,1}, \ldots, a_{\mu,n})$ of $A_\mu^\wedge$ are topologically nilpotent in $A_\mu^\wedge$. Thus we obtain a continuous R-algebra homomorphism

$$\Psi : R[x_1, \ldots, x_n]^\wedge \longrightarrow A_\mu^\wedge, \quad x_j \longmapsto a_{\mu,j}$$

In order to conclude what we want, we need to see if $\mathrm{Ker}(\Phi)$ is annihilated by Ψ. This may not be true, but we can achieve this after increasing μ. Indeed, since $R[x_1, \ldots, x_n]^\wedge$ is Noetherian, we can choose generators $g_1, \ldots, g_l$ of the ideal $\mathrm{Ker}(\Phi)$. Then we see that

$$\Psi(g_1), \ldots, \Psi(g_l) \in \mathrm{Ker}(A_\mu^\wedge \to C_\mu^\wedge) = \mathrm{Ker}(A_\mu \to C_\mu)$$

map to zero in $\mathrm{Ker}(A \to C) = \operatorname{colim} \mathrm{Ker}(A_\mu \to C_\mu)$. Hence increasing μ as before we get the desired result.

Proof of surjectivity: mopping up. The continuous ring homomorphism $B \to (A_\mu)^\wedge$ constructed above determines a morphism $\hat{x}_\mu : V_{\mu,/Z_\mu} \to W$. The compatibility of $\hat{x}_\mu$ and u'_μ follows from the fact that the ring map $B \to (A_\mu)^\wedge$ is by construction compatible with the ring map $A_\mu \to C_\mu$. In fact, the compatibility will be witnessed by the proper morphism $V'_\mu \to V_\mu$ and the morphisms x'_μ and $\hat{x}'_\mu = x'_{\mu,/Z_\mu}$ we used in the construction. This finishes the proof. □